Monitoring

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CloudWatch

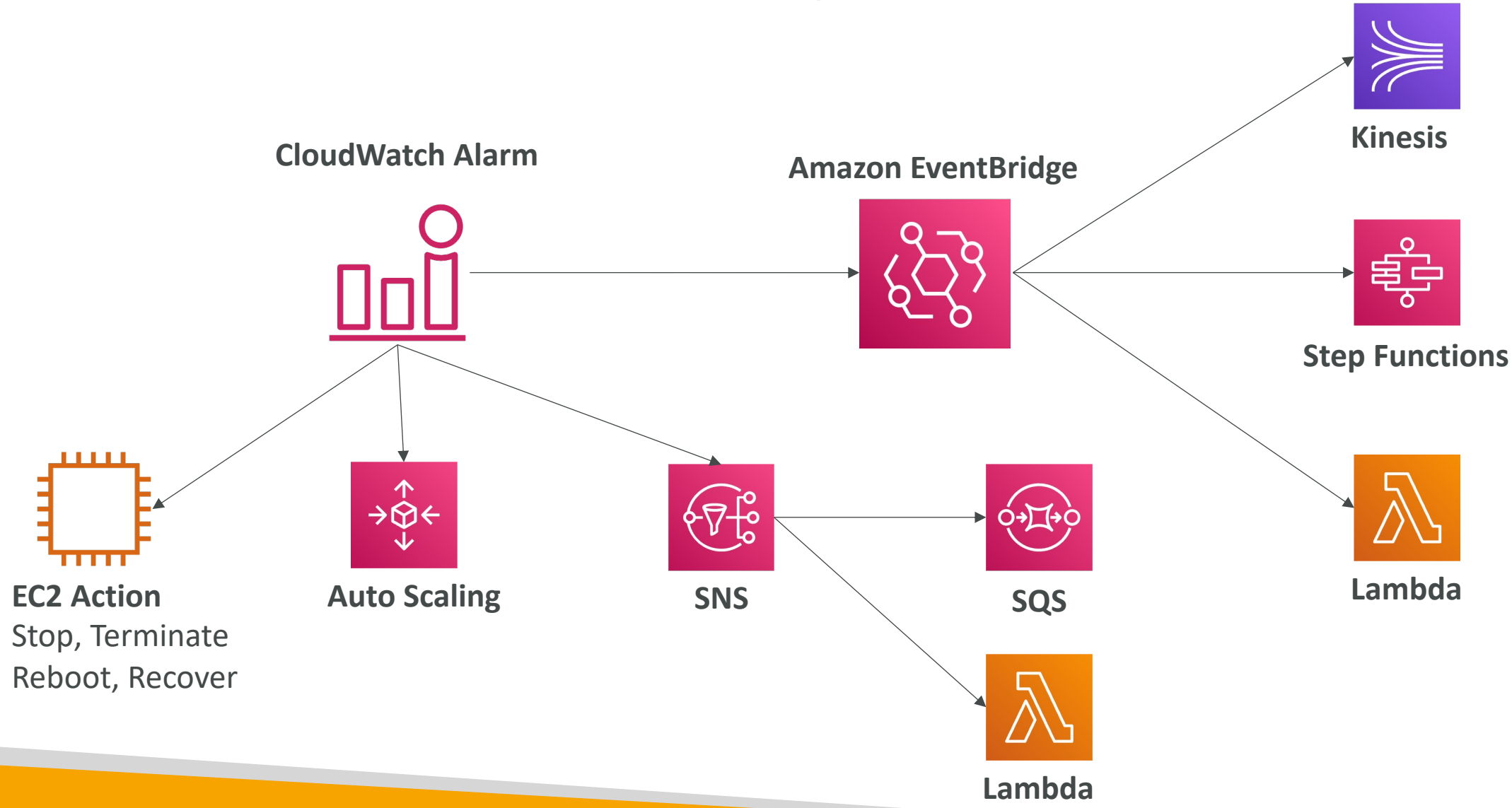


- CloudWatch Metrics
  - Provided by many AWS services
  - EC2 standard: 5 minutes, detailed monitoring: 1 minute
  - EC2 RAM is not a built-in metric
  - Can create custom metrics: standard resolution 1 minute, high resolution 1 sec
- CloudWatch Alarms
  - Can trigger actions: EC2 action (reboot, stop, terminate, recover), Auto Scaling, SNS
  - Alarm events can be intercepted by Amazon EventBridge
- CloudWatch Dashboards
  - Display metrics and alarms
  - Can show metrics of multiple regions

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CloudWatch Alarms integrations

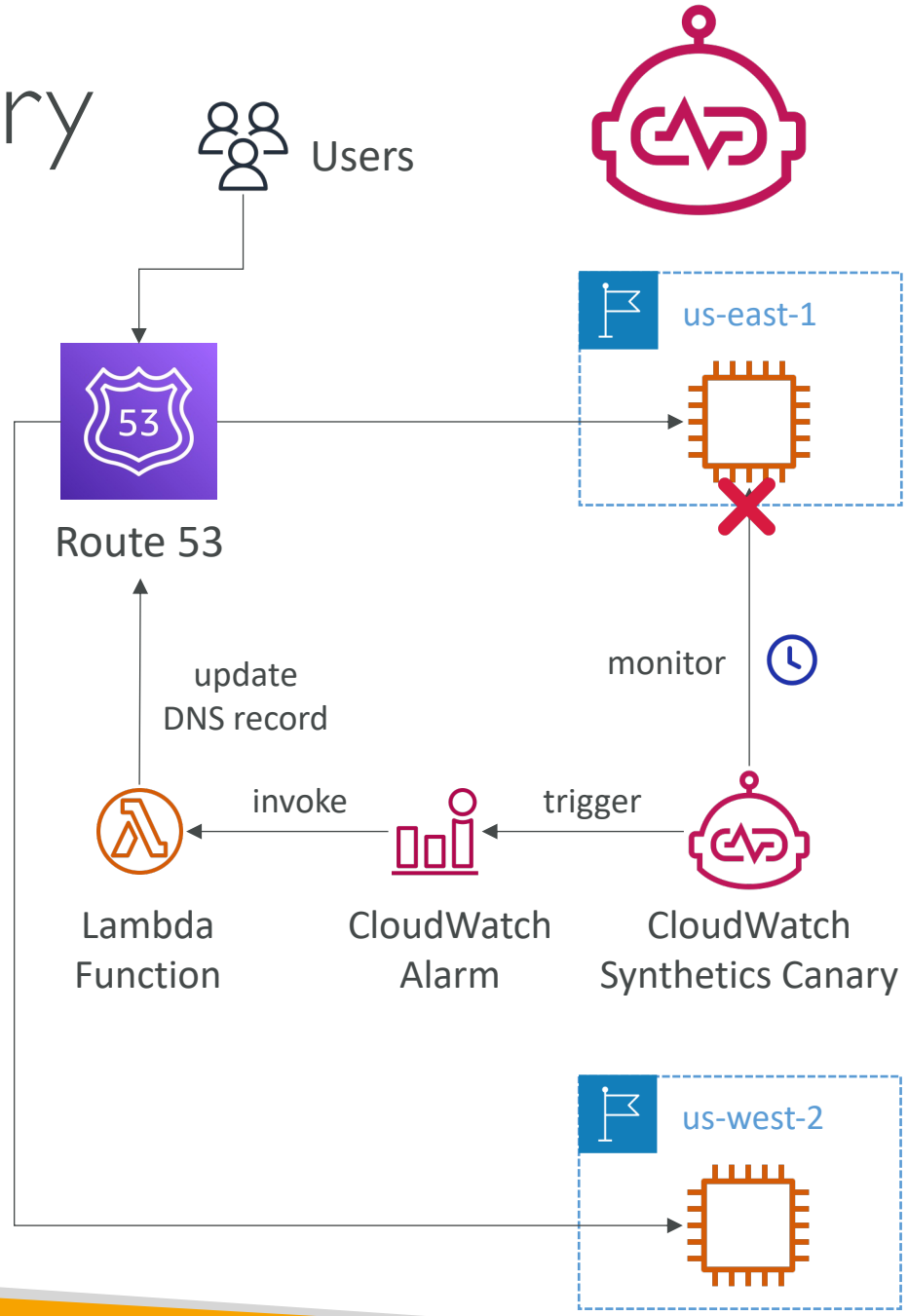


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CloudWatch Synthetics Canary

- Configurable script that monitor your APIs, URLs, Websites, …
- Reproduce what your customers do programmatically to find issues before customers are impacted
- Checks the availability and latency of your endpoints and can store load time data and screenshots of the UI
- Integration with CloudWatch Alarms
- Scripts written in Node.js or Python
- Programmatic access to a headless Google Chrome browser
- Can run once or on a regular schedule



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CloudWatch Synthetics Canary Blueprints

- **Heartbeat Monitor** – load URL, store screenshot and an HTTP archive file
- **API Canary** – test basic read and write functions of REST APIs
- **Broken Link Checker** – check all links inside the URL that you are testing
- **Visual Monitoring** – compare a screenshot taken during a canary run with a baseline screenshot
- **Canary Recorder** – used with CloudWatch Synthetics Recorder (record your actions on a website and automatically generates a script for that)
- **GUI Workflow Builder** – verifies that actions can be taken on your webpage (e.g., test a webpage with a login form)

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AWS CloudWatch Logs - Sources

- SDK, CloudWatch Logs Agent, CloudWatch Unified Agent
- Elastic Beanstalk: collection of logs from application
- ECS: collection from containers
- AWS Lambda: collection from function logs
- VPC Flow Logs: VPC specific logs
- API Gateway
- CloudTrail based on filter
- CloudWatch log agents: for example on EC2 machines
- Route53: Log DNS queries

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CloudWatch Logs

- **Log groups:** arbitrary name, usually representing an application
- **Log stream:** instances within application / log files / containers
- Can define log expiration policies (never expire, 30 days, etc..)
- Optional KMS encryption
- **CloudWatch Logs can send logs to:**
  - Amazon S3 (exports)
  - Kinesis Data Streams
  - Kinesis Data Firehose
  - AWS Lambda
  - ElasticSearch

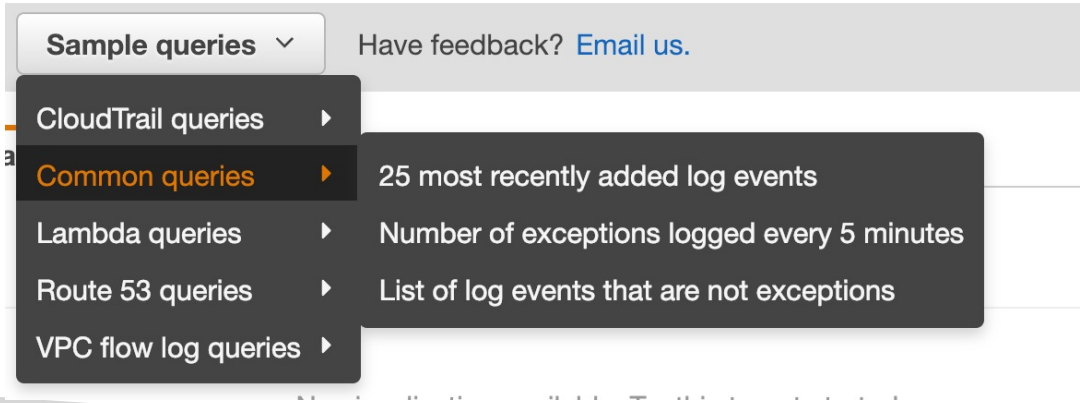
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CloudWatch Logs Metric Filter & Insights

- CloudWatch Logs can use filter expressions
  - For example, find a specific IP inside of a log
  - Or count occurrences of "ERROR" in your logs
  - Metric filters can be used to trigger alarms

- CloudWatch Logs Insights can be used to query logs and add queries to CloudWatch Dashboards



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CloudWatch Logs – S3 Export



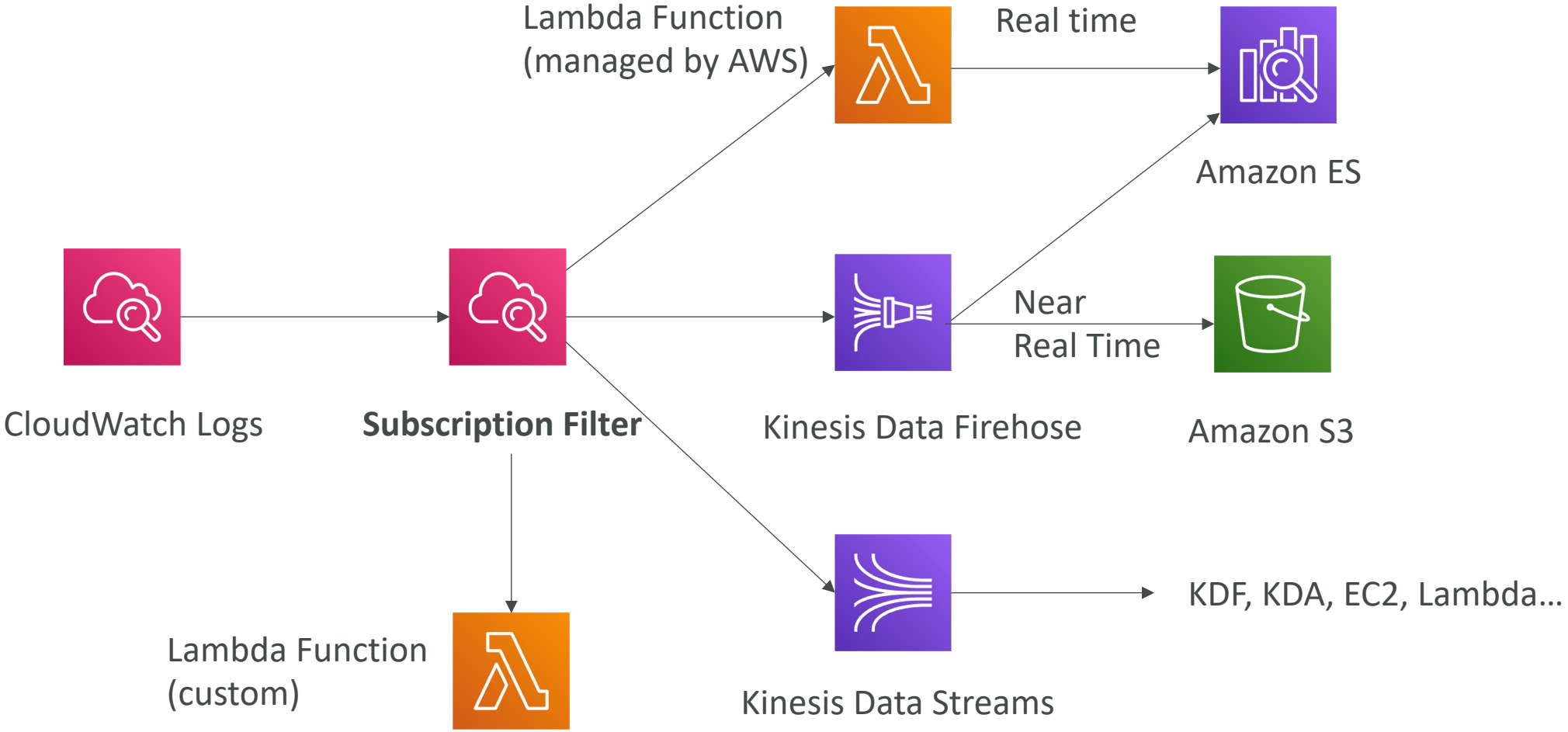
- S3 buckets must be encrypted with AES-256 (SSE-S3) or SSE-KMS
- Log data can take **up to 12 hours** to become available for export
- The API call is **CreateExportTask**

- Not near-real time or real-time… use Logs Subscriptions instead

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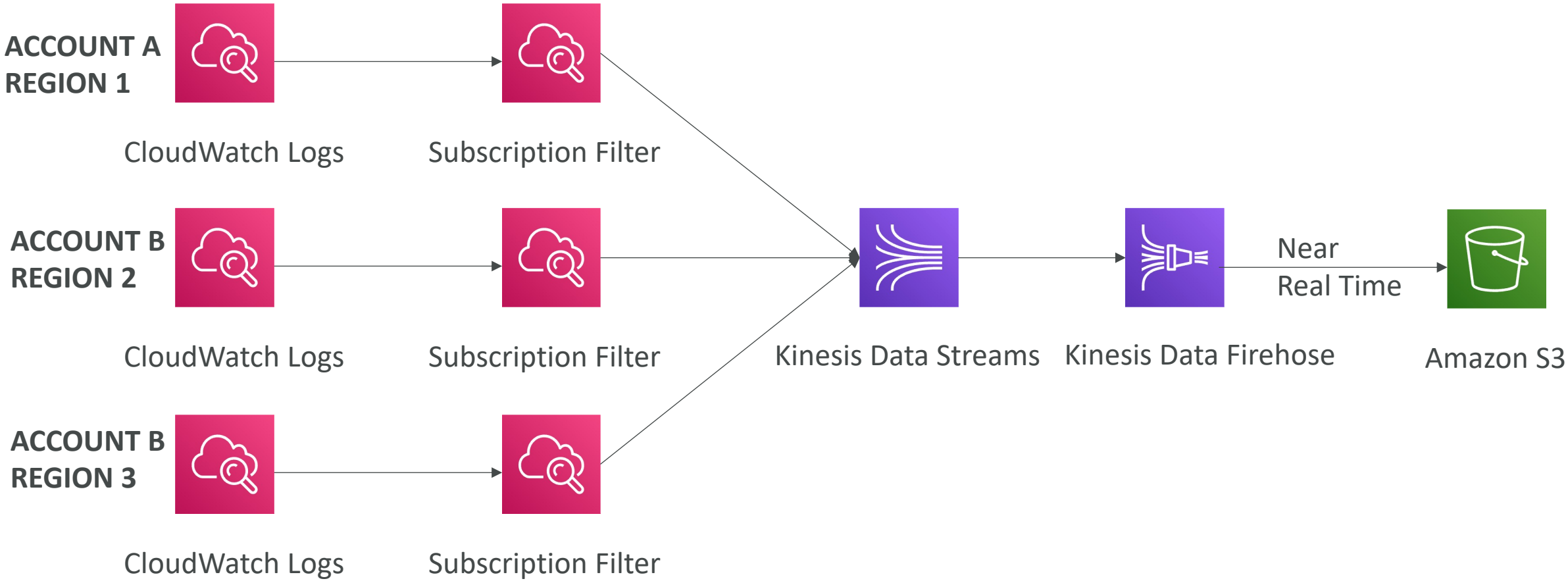
CloudWatch Logs Subscriptions



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CloudWatch Logs Aggregation Multi-Account & Multi Region



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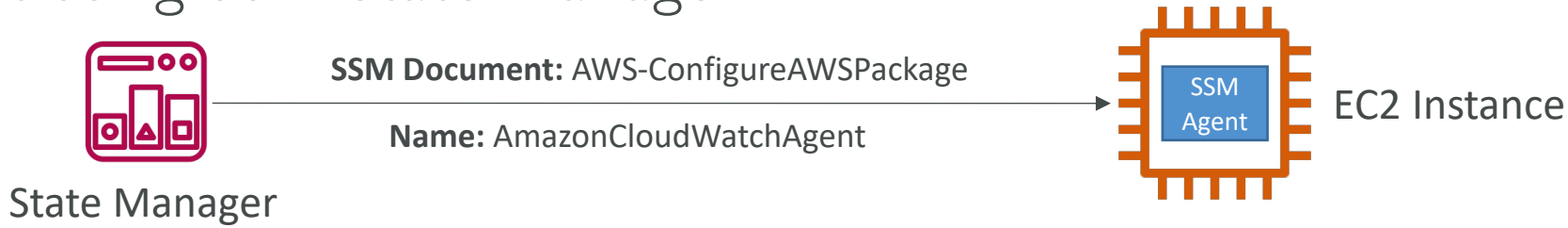
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CloudWatch Agent – Integration with SSM

- Install CW Agent using SSM Run Command



- Install CW Agent using SSM State Manager



- Configure CW Agent by storing config. in SSM Parameter Store



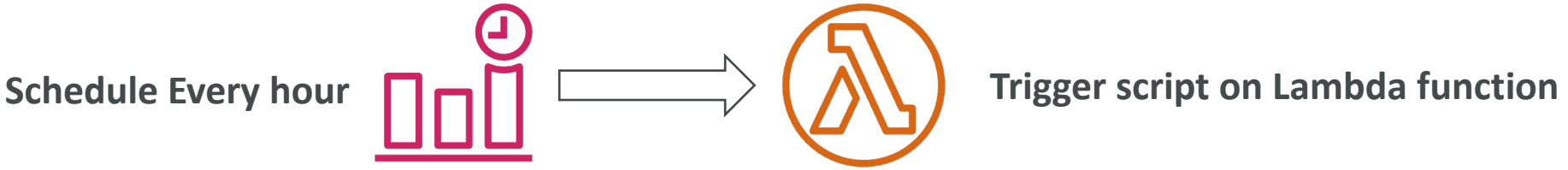
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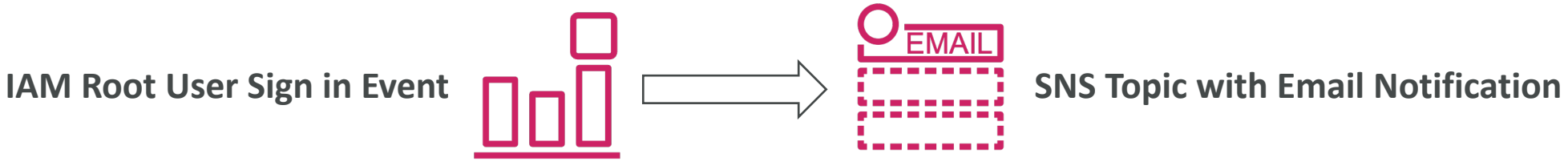
Amazon EventBridge (formerly CloudWatch Events)



- Schedule: Cron jobs (scheduled scripts)



- Event Pattern: Event rules to react to a service doing something

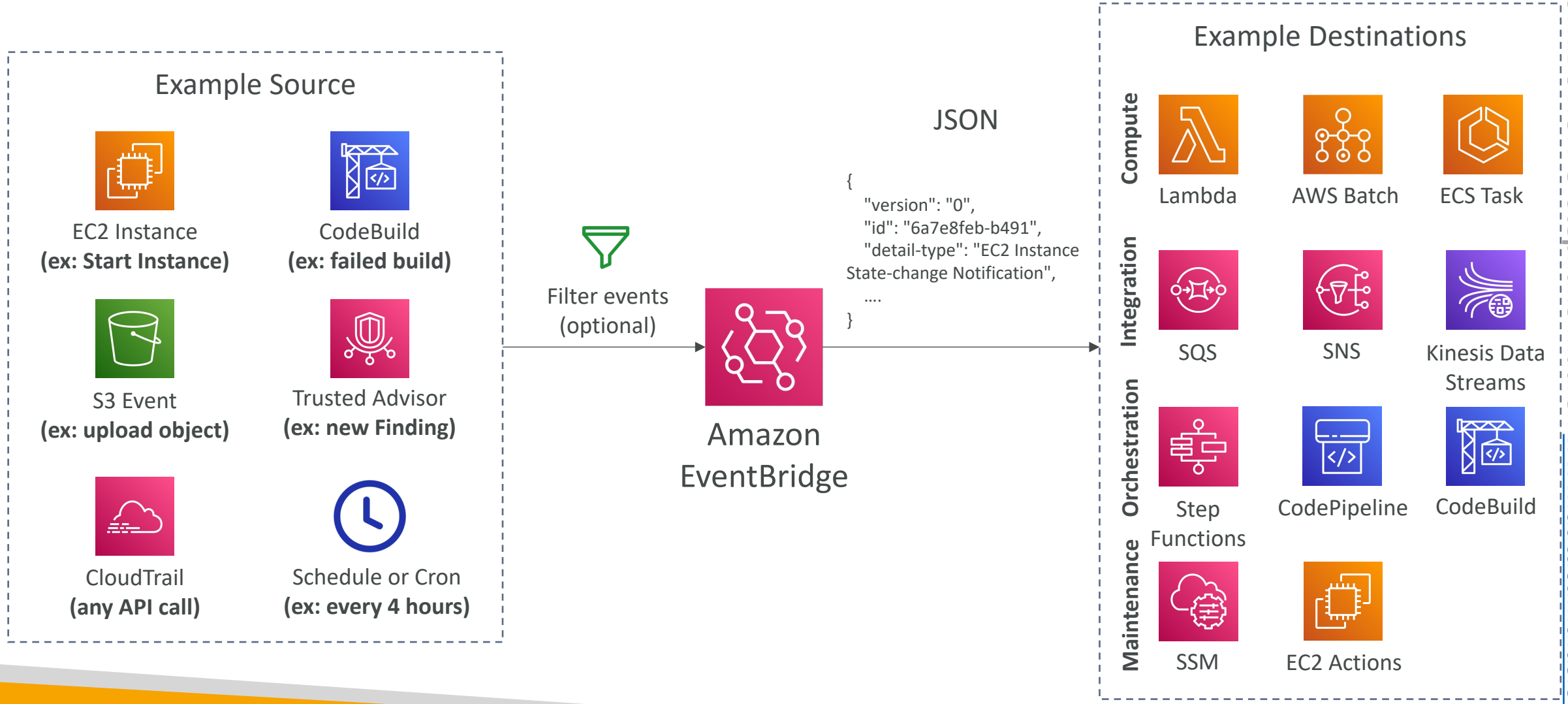


- Trigger Lambda functions, send SQS/SNS messages…

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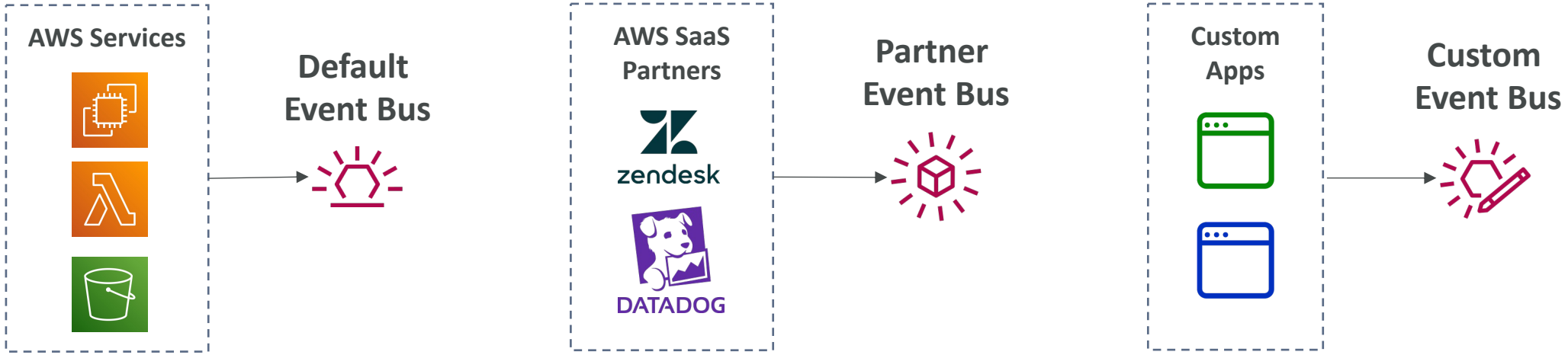
Amazon EventBridge Rules



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Amazon EventBridge



- Event buses can be accessed by other AWS accounts using Resource-based Policies

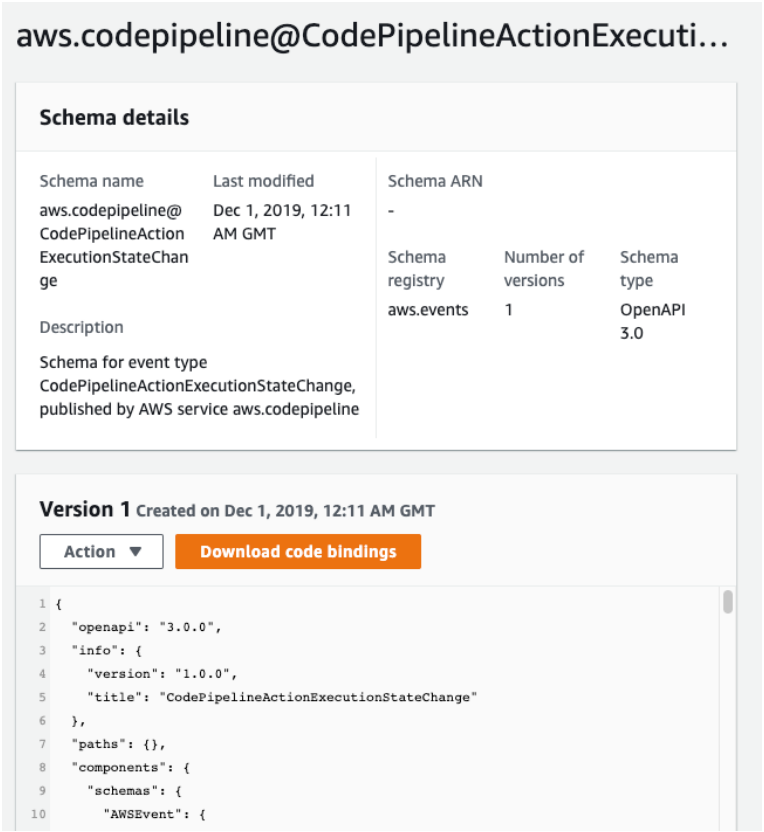
- You can **archive events** (all/filter) sent to an event bus (indefinitely or set period)
- Ability to **replay archived events**

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Amazon EventBridge – Schema Registry

- EventBridge can analyze the events in your bus and infer the **schema**
- The **Schema Registry** allows you to generate code for your application, that will know in advance how data is structured in the event bus
- Schema can be versioned



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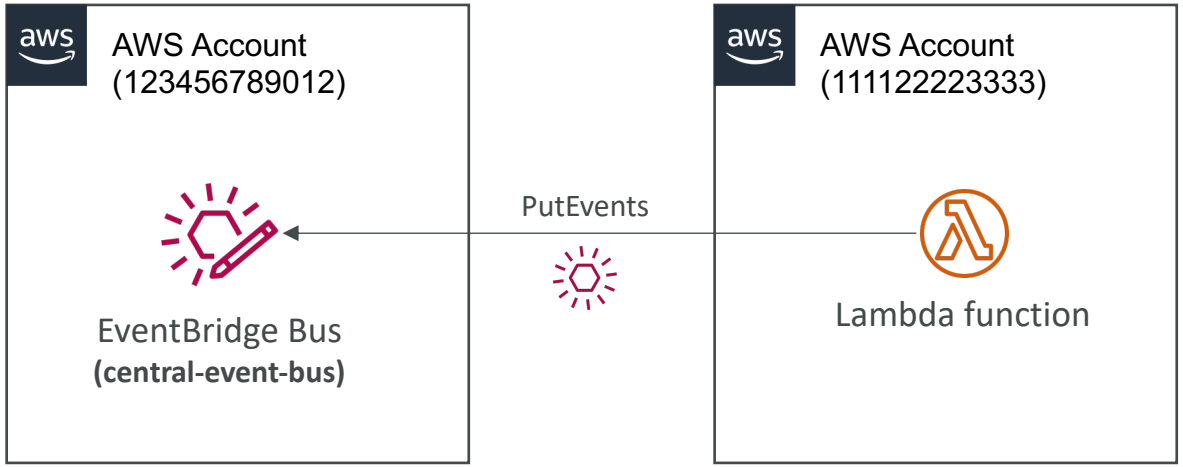
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Amazon EventBridge – Resource-based Policy

- Manage permissions for a specific Event Bus
- Example: allow/deny events from another AWS account or AWS region
- Use case: aggregate all events from your AWS Organization in a single AWS account or AWS region

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "events:PutEvents",
            "Principal": { "AWS": "111122223333" },
            "Resource": "arn:aws:events:us-east-1:123456789012:
event-bus/central-event-bus"
        }
    ]
}
```

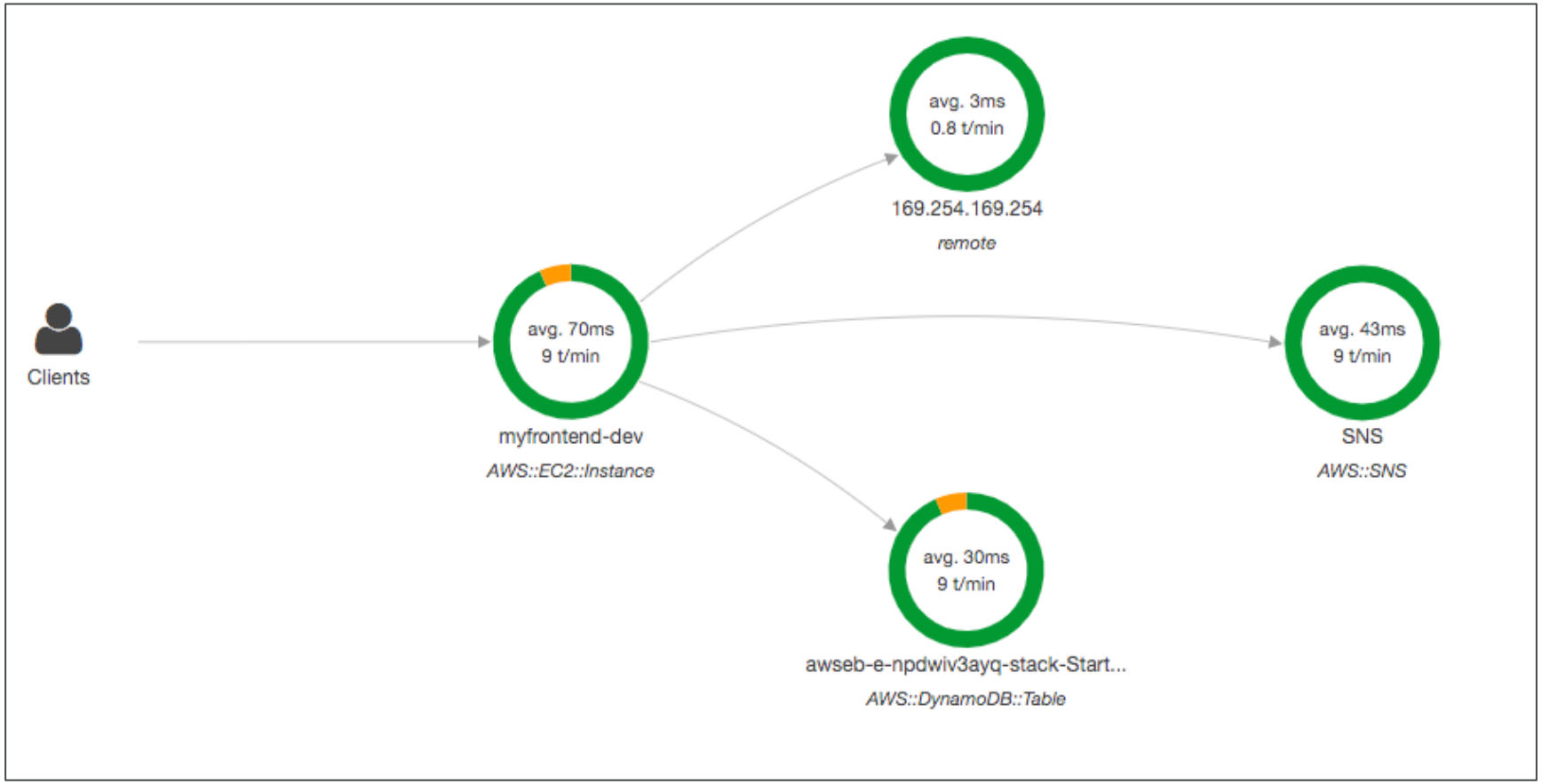
Allow **events** from another AWS account



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AWS X-Ray
Visual analysis of our applications



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X-Ray

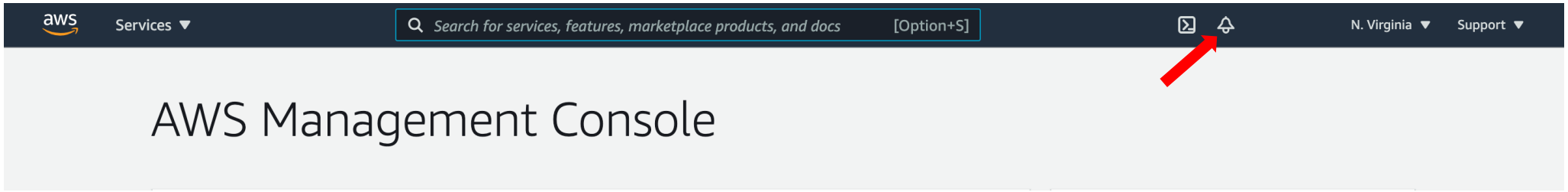


- Tracing requests across your microservices (distributed systems)
- Integrations with:
  - EC2 – install the X-Ray agent
  - ECS – install the X-Ray agent or Docker container
  - Lambda
  - Beanstalk - agent is automatically installed
  - API Gateway – helpful to debug errors (such as 504)
- The X-Ray agent or services need IAM permissions to X-Ray

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AWS Personal Health Dashboard



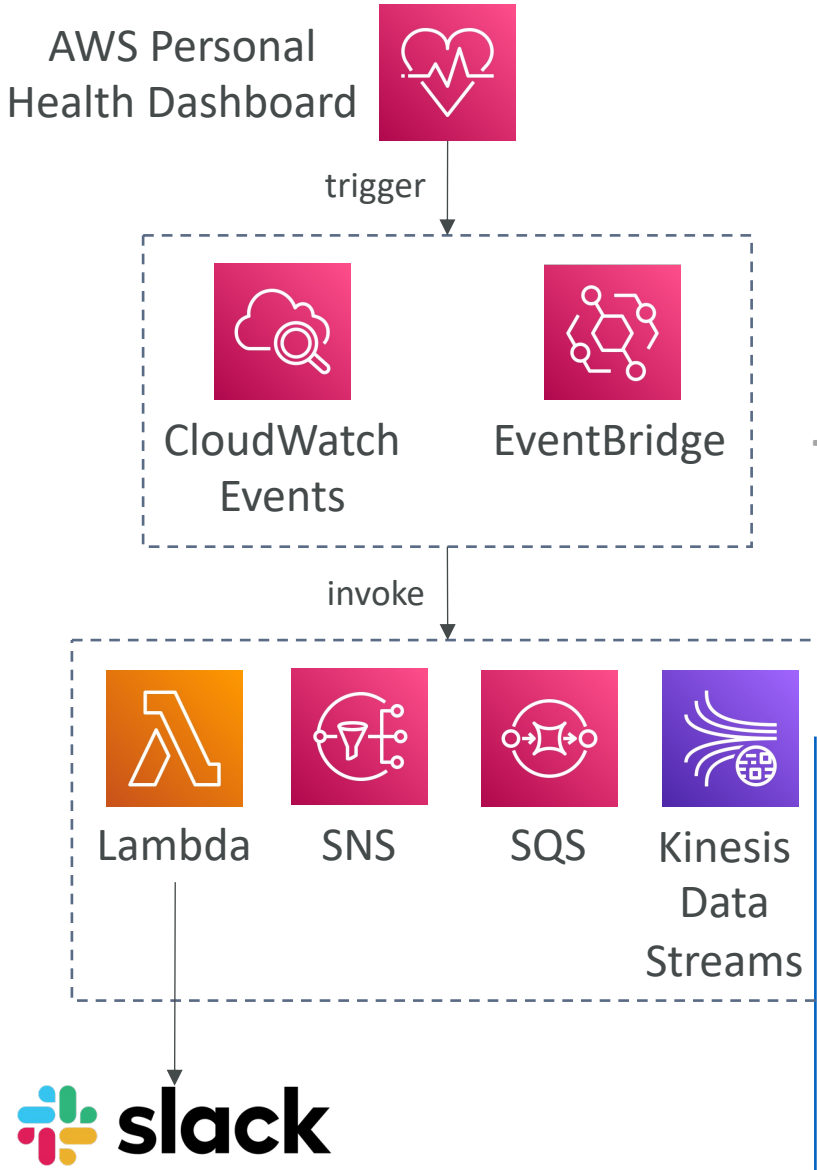
- Global service
- Show how AWS outages directly impact you
- Show the impact on your resources
- List issues and actions you can do to remediate them
- Will show **maintenance** events from AWS
- Programmatically accessible through the **AWS Health API**
- Aggregations across multiple accounts of an **AWS Organization**
- https://phd.aws.amazon.com/

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Health Event Notifications

- Use EventBridge (CloudWatch Events) to react to changes for AWS Health events in your AWS account
- Example: receive email notifications when EC2 instances in your AWS account are scheduled for updates
- Can't be used to return public events from the Service Health Dashboard
- Use cases: send notifications, capture event information, take corrective action, …



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